



FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

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CORE SKILLS

Languages: C++, C#, Python, HLSL/GLSL

Engines: Unity (URP), Unreal Engine 5, Custom OpenGL Frameworks

Software: Blender, DaVinci Resolve, Jira, Git

Workflows: Project Management, Tech Direction, Agile Product Strategy

EDUCATION

M.Sc. Cinema & Media Engineering

Politecnico di Torino | 2024 - 2026
Expected Grade: 110/110 Cum Laude (Dec 2026)

B.Sc. Cinema & Media Engineering

Politecnico di Torino | 2021 - 2024
Final Grade: 100/110

PERSONAL PROJECTS

Smart Cable Gen (Blender Tool)

Python / Pipeline Tooling
Procedural multi-object cable authoring add-on with distance-based catenary physics and auto-mesh export for game engines.

Graphics Lab (Custom C++ Engine)

Engine Programmer
OpenGL 3D rendering framework built from scratch with Deferred Shading, PBR, and Silhouette Parallax Occlusion Mapping (SPOM).

UE5 Tech Art Pipeline

PCG & Niagara Workflows
Procedural PCG biomes, Niagara homing projectiles, and MetaHuman custom groom collision physics.

Awaken Asylum & Nihil

Lead Technical Artist
Immersive VR experience and 2D Metroidvania prototype emphasizing technical production and custom shaders.

PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

WORK EXPERIENCE

XR Master Thesis - R&D; Graphics Programmer

03/2026 - Present

Lemons In The Room | Florence, Italy

- Engineered a real-time HLSL Raymarching framework for Signed Distance Fields (SDFs) in Unity (URP), enabling procedural, infinite-resolution topological morphing driven by biometric data.
- Developed an in-engine authoring tool with custom Shader Graph nodes and dynamic parameter routing, allowing artists and clinicians to visually compose and map live BLE biometric streams (ECG, HRV, breathing) to SDF morphology.
- Implemented Dynamic Bounding Volumes and adaptive step-size scaling to cut GPU fill-rate costs, guaranteeing a stable 72Hz framerate on standalone Meta Quest 3 hardware.

Technical Artist & Gameplay Programmer

01/2025 - Present

Level Up Lab | Turin, Italy

- Programmed core gameplay mechanics in C# (Unity) and designed the technical architecture for the 'Line Operator' project.
- Acted as Project Manager and Technical Producer, coordinating multidisciplinary teams of designers and artists to oversee artistic vision and technical viability.
- Developed custom HLSL Shaders and post-processing pipelines to establish the game's unique visual identity.

Teaching Assistant (GUI)

10/2024 - 01/2025

Politecnico di Torino | Turin, Italy

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created instructional content to improve comprehension.

Video Artist

09/2023 - 06/2024

RKH SRL | Turin, Italy

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.